ALLEN JOE A

VLSI & Semiconductor Engineer | Digital RTL, ASIC Physical Design & TCAD

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EDUCATION

Bachelor of Technology (B.Tech) in Electronics and VLSI Engineering *Expected 2028*

School of Electronics Engineering (SENSE), Vellore Institute of Technology (VIT), Chennai

- **Coursework:** Digital RTL Design, CMOS VLSI, ASIC Physical Design, Computer Architecture, Solid-State Devices, Verification.

TECHNICAL SKILLS

- **HDLs & Scripting:** Verilog-2001, SystemVerilog, Python, C/C++, Tcl, Bash.
- **ASIC & EDA Tools:** Cadence Genus, Cadence Innovus, Cadence Incisive, Cadence Voltus, OpenROAD, Yosys, ModelSim, QuestaSim.
- **TCAD & Physics:** Synopsys Sentaurus (SDE, SDevice, Inspect), TFETs, JLTFETs, Ferroelectric NC-FETs, Nanocavity Biosensors.
- **Architectures & Verification:** AMBA AXI4, APB, 4-Ch DMA, DFT / BIST (LFSR, MISR), Clock Gating, Bumpless Mode Transfer, Fixed-Point Q8.8.

VERIFIED VLSI & ASIC PROJECTS

4-Channel Adaptive Burst DMA Controller with Performance Monitoring *SystemVerilog | Questasim*

- Designed synthesizable SystemVerilog RTL featuring dynamic burst sizing (INCR4 to INCR16), round-robin arbitration, and non-intrusive 64-bit monitors.
- Verified across 6,266 automated self-checking assertion checks (0 errors); achieved 50% burst overhead reduction and 5.61% cycle latency reduction on 512-word transfers.

Gain-Scheduled All-Digital Phase-Locked Loop (GS-ADPLL) & ASIC Flow *Cadence Genus | Innovus*

- Designed dual-mode loop filter FSM switching dynamically from wideband acquisition (Gear 1) to low-jitter tracking (Gear 2, K=8).
- Cadence Incisive gate simulation confirmed 28-cycle fast lock acquisition and cycle-to-cycle steady-state jitter reduction from 38.4 ps down to 0.8–3.1 ps (91.9% reduction at K=8 knee; 87.5% tracking activity reduction). Cadence Innovus P&R; layout completed cleanly.

PS-02 Mode-Switching Flight Control PID — Silicon Dawn VLSI Hackathon Finalist *Cadence Genus | Innovus*

- Implemented cascaded quadcopter attitude/rate PID controller in synthesizable Verilog-2001 with signed Q8.8 arithmetic, anti-windup, and bumpless transfer.
- Synthesized in Cadence Genus and physically placed & routed in Cadence Innovus: achieved timing closure at 50 MHz (+0.016 ns slack) with 0 DRC and 0 connectivity violations across 446 × 444 µm core.

Low-Power 16-Bit MAC BIST Architecture with Clock Gating *Cadence Genus | Incisive*

- Synthesized 16-bit Multiply-Accumulate unit with integrated low-power BIST (LFSR pattern generator, MISR signature compaction, low-transition gating).
- Achieved 99.8980% stuck-at fault coverage across 1,960 injected faults in Cadence Incisive; verified zero setup/hold timing violations at 100 MHz in Genus 21.1.

FE-DGDM-JLTFET Nanocavity Biosensor Device Physics Modeling *Synopsys Sentaurus TCAD*

- Simulated 10nm negative-capacitance junctionless TFET combining HZO ferroelectric gate with inverted-T biomolecule cavity under Dr. Lakshmi B at SENSE.
- Extracted quantum drift-diffusion and Hurkx BTBT transport: Ion = 0.8040 mA, Ion/Ioff = 7.24 × 10■, and 18.14% sensitivity for MDA-MB-231 breast cancer cells.

GNN-T Edge-Placement Error Prediction for Computational Lithography *PyTorch | PyTorch Geometric*

- Built E(n)-equivariant graph neural network + Transformer predicting sub-resolution edge placement errors (EPE) from layout polygon topologies.
- Accelerated optical proximity correction (OPC) exploration by 14× over numerical lithography solvers with sub-nanometer prediction error.